

CHAPTER III

DESIGN AND METHODOLOGY

This chapter of the paper presents an overview of the proposed study's fundamental theoretical foundations. The research framework is guided by these foundations in order to give substantial results to the research questions at the conclusion of the study. It was also utilized to monitor the progression of the system's workflow.

Research Design

In this research, a quasi-experimental research design has been adopted, focusing explicitly on evaluating the behavioral impact of the StudyCircle system on its target users. Quasi-experimental research is employed to track and isolate cause-and-effect relationships without the utilization of randomized participant assignment. This design is highly relevant given the structured nature of academic environments where forming strictly randomized classes is logistically disruptive.

A Non-Equivalent Control Group Pre-Test/Post-Test Design is implemented within the context of this study. Intact student cohorts or sections are carefully designated into two distinct testing boundaries: the **Experimental Group** (who will study within the complete StudyCircle platform, experiencing automated WSM calculation loops, digital nudge alerts, and virtual pet tracking mechanisms) and the **Control Group** (who will carry out their learning sessions using standard, non-gamified focus methods). This methodology bridges theory and practice, offering both theoretical insights and practical solutions to the student community, aligning with the

research's overarching objective.

Research Methodology

Agile/ GDLC (Pre-production , Production, Post Production)

The researchers adopted the Agile Software Development Methodology in developing the StudyCircle system. Agile was selected because of its iterative, collaborative, and adaptive nature, which aligns with the dynamic requirements of educational technology systems and user-centered application development.

The Agile methodology allowed the proponents to continuously improve the StudyCircle system through multiple iterations involving planning, designing, development, testing, deployment, and review. Frequent feedback from potential users and stakeholders enabled the system to evolve according to actual user needs and usability requirements.

The methodology consists of the following phases:

Plan

The planning phase served as the foundation of the entire study. During this phase, the researchers identified the primary problems encountered by students in remote and hybrid learning environments, particularly cyberloafing, lack of accountability, procrastination, and declining productivity.

The proponents conducted literature reviews, gathered related studies, and analyzed existing productivity applications in order to define the scope and objectives of the proposed system. The researchers also identified the key features necessary for the application, such as:

- Multiplayer virtual study rooms;
- Structured study timers;
- Productivity scoring system;
- Gamified reward mechanics;
- Real-time collaboration features;
- Peer accountability mechanisms; and
- AI-assisted quizzes and productivity tools.

The Agile planning phase also involved defining the project requirements, identifying stakeholders, assigning development tasks, and preparing sprint objectives for every development iteration. Continuous communication among the researchers ensured that project goals remained aligned throughout the development lifecycle.

Design

Following the planning phase, the researchers proceeded to the design phase where the overall architecture, algorithm structure, database organization, and user interface of the system were conceptualized.

The design phase focused on ensuring that the StudyCircle application would provide a user-friendly, responsive, and engaging environment for students while maintaining reliable productivity tracking mechanisms.

Algorithm Design – Weighted Sum Model (WSM)

The primary algorithm utilized in the system is the Weighted Sum Model (WSM), which serves as the productivity evaluation mechanism of StudyCircle.

The Weighted Sum Model calculates the overall productivity score of a student based on five identified productivity criteria:

- Study Duration – 40%
- Task Completion – 25%
- Session Consistency – 15%
- Group Participation – 10%
- Peer/Self-Evaluation – 10%

The formula utilized in the system is:

Where:

- w_i represents the assigned weight of each criterion;
- x_i represents the normalized value of the criterion;
- n represents the total number of evaluation criteria.

The algorithm was selected because of its simplicity, transparency, deterministic output, and traceable computation process. It allows students to clearly understand how their productivity scores are computed while minimizing ambiguity in the evaluation process.

System Structure

The proposed system follows a client-server web architecture. The frontend interface interacts with the backend server through API requests and real-time socket communication.

The major components of the system include:

- User Authentication Module
- Study Room Management Module
- Productivity Scoring Module
- Gamification and Reward System
- AI Quiz Module
- Notification and Help Request Module
- Database Management System

The database stores user accounts, productivity scores, session logs, room data, reward inventories, and participation records.

Application Design

The StudyCircle interface was designed with emphasis on usability, simplicity, and responsiveness. The proponents designed the application to be compatible across desktop and mobile web browsers.

The user interface contains the following primary pages and modules:

- Login and Registration Interface
- Dashboard Page
- Virtual Study Room Interface
- Productivity Statistics Page
- Reward and Inventory System
- AI Quiz Interface

- User Profile and Leaderboards

The researchers also considered Cognitive Load Theory during interface design by minimizing unnecessary interface clutter and organizing content into clear visual sections.

Gamification elements such as virtual pets, avatars, badges, streak systems, and leaderboards were integrated to improve engagement and motivation among users.

Develop

During the development phase, the researchers implemented the designed system functionalities incrementally following Agile sprint cycles.

The StudyCircle system was developed using the following technologies:

- Python Flask for backend development;
- HTML5, CSS3, and JavaScript for frontend development;
- Bootstrap framework for responsive design;
- SQLite database for local data storage;
- WebSockets for real-time synchronization;
- Bcrypt for password hashing and account security.

Algorithm Development

The Weighted Sum Model algorithm was implemented within the backend server logic. User study activities were continuously tracked and stored in the database. The algorithm processes collected productivity metrics and generates real-time productivity scores.

The development process included:

- Data normalization;
- Weight assignment validation;
- Score computation functions;
- Real-time score updates;
- Reward trigger conditions.

The researchers also implemented validation mechanisms to ensure that productivity scores remained fair and resistant to manipulation.

Application Development

The application modules were developed incrementally during Agile sprints. Each sprint focused on specific functionalities such as:

- User authentication;
- Room creation and synchronization;
- Productivity monitoring;
- Gamification systems;
- AI-assisted quiz generation;
- Real-time notifications.

Continuous debugging and refinement were conducted throughout development. User feedback obtained during prototype demonstrations also guided improvements in system navigation, interface layout, and feature accessibility.

Test

After development, the StudyCircle system underwent rigorous testing procedures to ensure reliability, usability, functionality, and performance.

Algorithm Evaluation

The researchers evaluated the effectiveness of the Weighted Sum Model by analyzing the consistency and fairness of generated productivity scores across multiple simulated study sessions.

The evaluation focused on:

- Accuracy of score computation;
- Consistency of weighted outputs;
- Reliability of productivity tracking;
- Traceability of scoring criteria.

Multiple testing scenarios were conducted to verify whether the algorithm properly reflected user productivity behaviors according to the predefined evaluation weights.

System Testing

The StudyCircle application underwent both functional and non-functional testing.

Functional Testing

The following functionalities were tested:

- User registration and login;
- Study room creation and joining;
- Real-time timer synchronization;
- AI quiz generation;
- Productivity score computation;
- Reward and inventory systems;
- Help request notifications.

Test cases were conducted to verify whether each module behaved according to expected outputs.

Non-Functional Testing

The system was also evaluated according to non-functional quality attributes such as:

- Security;
- Reliability;
- Usability;
- Performance efficiency;
- Compatibility.

Password encryption using Bcrypt was tested to ensure account security. Latency testing was also conducted to verify that room synchronization remained below the targeted 200ms response delay.

Usability testing was performed through user evaluations using ISO 25010 software quality metrics.

Deploy

Following successful testing and refinement, the StudyCircle system was deployed for limited real-world usage and demonstration.

The researchers prepared the application for deployment using a Flask-based hosting environment. The deployment process involved:

- Server configuration;
- Database migration;
- User accessibility testing;
- Real-time synchronization validation;
- Security verification.

The deployment phase allowed actual users to interact with the system in simulated remote learning scenarios.

Review

The final phase of the Agile methodology involved reviewing the overall development process and evaluating the effectiveness of the StudyCircle system.

The researchers conducted retrospective analyses after every development sprint in order to identify:

- Development challenges;
- System limitations;
- User feedback trends;

- Areas for future improvement.

Feedback gathered from users highlighted several recommendations, including:

- Mobile application integration;
- Additional customization features;
- Improved AI-assisted productivity tools;
- Enhanced collaboration systems.

The Agile review process enabled the proponents to continuously refine the application and ensure that the final system remained aligned with the study's objectives and user needs.